signal.sci (C:\Users\LENOVO\Documents\signal.sci) - SciNotes

signal.sci

```
X_fft = DIT_FFT(x);

// 5. Magnitude Spectrum
magX = abs(X_fft);

// 6. Plot Input Signal - BLUE
scf(0);
clf();

subplot(2,1,1);
plot(n, x, "b");
xgrid();
xtitle("Input Signal", "Sample Index (n)", "Amplitude");

// 7. Plot Magnitude Spectrum - RED
subplot(2,1,2);
plot2d3(n, magX, style=5);
xgrid();
xtitle("Magnitude Spectrum using DIT FFT", ...
       "Frequency Index (k)", "|X(k)|");

// 8. Display FFT results
disp("Input Signal:");
disp(x);

disp("DIT FFT Output:");
disp(X_fft);

disp("Magnitude Spectrum:");
disp(magX);

```

